

EMPIRICAL LIKELIHOOD FOR PROBABILITY DENSITY FUNCTIONS UNDER ϕ -MIXING SAMPLES

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Summary: This paper studies the construction of confidence intervals for a probability density function under a ϕ -mixing sample by using the blockwise technique. It is shown that the blockwise empirical likelihood (EL) ratio statistic is asymptotically χ^2 distributed, which is used to obtain EL-based confidence interval for the probability density function.

1. Introduction

Let $\{\eta_i, 1 \geq 1\}$ be a random variable sequence and $\mathcal{F}_s^t$ denote the σ -algebra generated by $\{\eta_i, s \leq i \leq t\}$ for $s \leq t$. Random variables $\{\eta_i, 1 \geq 1\}$ are said to be ϕ -mixing, if

$$\phi(n) = \sup_{k \geq 1} \left\{ \left| \frac{P(AB)}{P(A)} - P(B) \right| : A \in \mathcal{F}_1^k, B \in \mathcal{F}_{k+n}^\infty, P(A) > 0 \right\} \downarrow 0$$

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as $n \rightarrow \infty$, where $\phi(n)$ is called as ϕ -mixing coefficient.

The ϕ -mixing condition was introduced by Ibragimov (1959), and was also studied by Cogburn (1960). The concept of ϕ -mixing has been used extensively in the time series literature as measures of weak dependence. Bradley (2005) made a good survey on the ϕ -mixing condition as well as other commonly used mixing conditions. The convergence of the sums of ϕ -mixing random variables were studied extensively by Utev (1990), Chen (1991), Herrndorf (1983), Peligrad (1985), Sen (1971), Sen (1974), Shao (1993) and Wang, Hu, Yan and Yang (2009), among others.

The kernel estimation method of a probability density function (p.d.f.) was first proposed by Rosenblatt (1956). Under independent samples, the properties of the kernel estimate of a p.d.f. have been studied extensively. See Rosenblatt (1956), Parzen (1962), Nadaraja (1965), Singh (1977), Silverman (1986), among others. Robinson (1983) studied the asymptotic normality of the kernel estimators of a p.d.f. under α -mixing samples. Lu (2001) investigated the asymptotic normality of the kernel estimators of a p.d.f. under more general dependent samples. Lin (1983) studied the asymptotic normality of the kernel estimate of a p.d.f. under ϕ -mixing observations. However, these results can not be directly used to construct the confidence intervals for the p.d.f. as there are unknown asymptotic variances involved, which could be used to construct the normal-approximation-based confidence intervals for the p.d.f. after the unknown asymptotic variances are estimated. Our aim is to construct the empirical likelihood (EL) confidence intervals for a p.d.f. under a ϕ -mixing sample by using the blockwise technique. The estimation of unknown asymptotic variances under dependence is much more involved than independent cases in the construction of the normal-approximation-based confidence intervals. The fact that the EL intervals are obtained by contouring the likelihood ratio without a secondary variance estimation is particularly advantageous for dependent data.

The EL method to construct confidence intervals, proposed by Owen (1988), Owen (1990), has many advantages over its counterparts like the normal-approximation-based method and the bootstrap method (e.g., Hall and La Scala, 1990; Hall, 1992). Chen (1996) developed the EL method in the construction of a p.d.f. under an independent sample.

Kitamura (1997) first proposed blockwise EL method to construct confidence intervals for parameters with mixing samples. Chen and Wong (2009) developed blockwise EL method to construct confidence intervals for quantiles with α -mixing samples. Qin, Li, and Lei (2011) used the blockwise EL method to construct confidence intervals for a p.d.f. with a strong stationary negatively associated sample. Our motivation to study the construction of the EL-based confidence intervals for a p.d.f. under ϕ -mixing data is summarized as follows. (1) The strong mixing condition in Kitamura (1997) contains ϕ -mixing. However, the construction of the EL-based confidence intervals in Kitamura (1997) does not involve a p.d.f. (2) There is a lack of research on the construction of the EL-based confidence intervals for a p.d.f. under ϕ -mixing observations. (3) It is possible to generate the results in this paper to broader mixing conditions, which is left for our future study.

The rest of the paper is organized as follows. The main results of this paper are presented in Section 2. Some lemmas to prove the main results and the proof of the main results are presented in Section 3. Some concluding remarks are given in Section 4.

2. Main results

Let $X_1, \dots, X_n, \dots$ be a strictly stationary ϕ -mixing sequence with mixing coefficient $\phi(\cdot)$ from a population $X \in R$ and $f(\cdot)$ be the p.d.f. of X . The kernel density estimator for $f(x)$ at a given $x \in R$ is

$$f_n(x) = (nh)^{-1} \sum_{i=1}^n K_h(x - X_i),$$

where $K_h(u) = K(u/h)$, K is a kernel function, and $h = h_n$ are bandwidths.

To obtain the EL confidence intervals for $\theta = f(x)$ at a given $x \in R$, we employ the small-block and large-block arguments for the score functions $h^{-1}K_h(x - X_i) - \theta$, $1 \leq i \leq n$, proposed in Chen (1996), as follows. Let $e_i = h^{-1}K_h(x - X_i) - \theta$, $1 \leq i \leq n$, $k = [n/(p+q)]$, where $[t]$ denotes the integral part of t , and $p = p(n)$ and $q = q(n)$ are positive integers satisfying $p+q \leq n$. Put

$$\omega_{n,2m-1}(\theta) = \sum_{i=r_m}^{r_m+p-1} e_i, \omega_{n,2m}(\theta) = \sum_{i=l_m}^{l_m+q-1} e_i, \omega_{n,2k+1}(\theta) = \sum_{i=k(p+q)+1}^n e_i,$$

where $r_m = (m-1)(p+q) + 1$, $l_m = (m-1)(p+q) + p + 1$, $m = 1, \dots, k$.

We consider the following blockwise EL ratio

$$R(\theta) = \sup_{p_1, \dots, p_{2k+1}} \left\{ \prod_{j=1}^{2k+1} (2k+1)p_j \mid \sum_{j=1}^{2k+1} p_j = 1, p_j \geq 0, \sum_{j=1}^{2k+1} p_j \omega_{nj}(\theta) = 0 \right\}.$$

It is easy to obtain the $(-2\log)$ blockwise EL ratio statistic

$$\ell(\theta) = 2 \sum_{j=1}^{2k+1} \log \{1 + \lambda(\theta) \omega_{nj}(\theta)\}, \quad (1)$$

where $\lambda(\theta)$ is determined by

$$\frac{1}{2k+1} \sum_{j=1}^{2k+1} \frac{\omega_{nj}(\theta)}{1 + \lambda(\theta) \omega_{nj}(\theta)} = 0. \quad (2)$$

To obtain the asymptotical distribution of $\ell(\beta)$, we need the following assumptions.

Assumptions.

(A1) (i) The r.v.s $\{X_i, i \geq 1\}$ form a strictly stationary sequence, and let f be the one-dimensional marginal p.d.f. (with respect to Lebesgue measure).

(ii) The $\{X_i, i \geq 1\}$ are ϕ -mixing with $\sum_{n=1}^{\infty} \phi^{1/2}(n) < \infty$.

(iii) If $f_{1,j}$ is the joint p.d.f. of the r.v.s X_1, X_{j+1} , then $|f_{1,j}(u, v) - f(u)f(v)| \leq C$ for all $u, v \in R$ and $j > 1$.

(iv) $f(\cdot)$ has continuous derivatives up to the r th in a neighborhood of x , and $f(x) > 0$, where $r \geq 2$ is some integer.

(A2) The function K is bounded, has compact support and satisfies

$$\int u^j K(u) du = \begin{cases} 1 & \text{if } j = 0, \\ 0 & \text{if } 1 \leq j \leq r-1, \end{cases}$$

with r appearing in (A1)(iv).

(A3) Let p, q and k be as described above, and let $h > 0$ be bandwidths, which satisfy

(i) $pk/n \rightarrow 1$. (ii) $ph \rightarrow 0$. (iii) $q \rightarrow \infty$, $k \rightarrow \infty$ and $p^2/(nh) \rightarrow 0$. (iv) $k\phi(q) \rightarrow 0$.

(v) $nh^{2r+1} \rightarrow 0$ with r appearing in (A1)(iv).

Remark 1. Let $K(u) = (1 - u^2)^2 (\sum_{j=0}^r a_j u^j) I(|u| \leq 1)$, where I is the indicator function. By putting this kernel into Assumption (A2)(i) and solving linear equations to obtain $a_j, 0 \leq j \leq r$, we can see that K satisfies Assumption (A2).

Remark 2. $pk/n \rightarrow 1$ implies $qk/n \rightarrow 0$ and $q/p \rightarrow 0$. $p^2/(nh) \rightarrow 0$ implies $nh \rightarrow \infty$. $ph \rightarrow 0$ implies $qh \rightarrow 0$ and $h \rightarrow 0$. As $0 \leq n - k(p + q) \leq p + q$, we can obtain that $0 \leq n - k(p + q) \leq Cp$ from $q/p \rightarrow 0$. These facts will be used throughout this article. However, no explicit mentioning will be made.

Remark 3. Suppose that $\phi(n) = O(n^{-2-\delta})$ for some $\delta > 0$. Let $h = c_1 n^{-\beta_1}$, $p = [c_2 n^{\beta_2}]$ and $q = [c_3 n^{\beta_3}]$ for some constants $c_j, 1 \leq j \leq 3$, where $0 < \beta_3 < \beta_2 < \beta_1 < 1$. If we first choose β_2 and β_3 to satisfy $2/\{3(2+\delta)\} < \beta_3 < \beta_2 < 1/3$ and then take β_1 such that $\min\{\beta_2, 1/(2r+1)\} < \beta_1 < 1 - 2\beta_2$, then (A3) is satisfied.

We now state the main results in this paper.

THEOREM 1 Suppose that Assumptions (A1) to (A3) are satisfied. Then, as $n \rightarrow \infty$,

$$\ell(\theta) \xrightarrow{d} \chi_1^2,$$

where χ_1^2 is a chi-squared distributed random variable with one degree of freedom.

Let z_α satisfy $P(\chi_1^2 \leq z_\alpha) = 1 - \alpha$ with $0 < \alpha < 1$. It follows from Theorem 1 that an EL-based confidence region for $\theta = f(x)$ with asymptotically correct coverage probability $1 - \alpha$ can be constructed as

$$\{\theta : \ell(\theta) \leq z_\alpha\}.$$

3. Lemmas and the proof of main results

We use C to denote a positive constant independent of n , which may take a different value for each appearance. To prove our main results, we need the following lemma.

LEMMA 1 Let $\{\eta_j : j \geq 1\}$ be a ϕ -mixing sequence with mixing coefficient $\phi(n)$ and $\mathcal{F}_s^t$ denote the σ -algebra generated by $\{\eta_i, s \leq i \leq t\}$ for $s \leq t$. Suppose that $\{f_j(\cdot), j \geq 1\}$ are all measurable functions and that $\{\xi_i, i \geq 1\}$ are $\mathcal{F}_{i_1}^{j_1}, \dots, \mathcal{F}_{i_n}^{j_n}, \dots$, measurable, respectively with $1 \leq i_1 < j_1 < i_2 < \dots < j_n < \dots$. Then, $\{f_j(\eta_j) : j \geq 1\}$ is a ϕ -mixing sequence with mixing coefficient $\phi_1(n) \leq \phi(n)$ and $\{\xi_j : j \geq 1\}$ is a ϕ -mixing sequence with mixing coefficient $\phi_2(n) \leq \phi(n)$.

Proof. This result can be directly proved by the definition of ϕ -mixing random variables.

LEMMA 2 Let $\{\eta_j : j \geq 1\}$ be a ϕ -mixing sequence with mixing coefficient $\phi(n)$ and $\mathcal{F}_s^t$ denote the σ -algebra generated by $\{\eta_i, s \leq i \leq t\}$ for $s \leq t$. Suppose that $\xi_1 \in \mathcal{F}_1^k$ and $\xi_2 \in \mathcal{F}_{k+n}^\infty$ for some $k \geq 1$, $E|\xi_1|^{p_1} < \infty$ and $E|\xi_2|^{p_2} < \infty$ for some p_1 and q_1 with $p_1 > 1, q_1 > 1, 1/p_1 + 1/q_1 = 1$. Then

$$|E(\xi_1 \xi_2) - (E\xi_1)(E\xi_2)| \leq 2\phi^{1/p_1}(n)(E|\xi_1|^{p_1})^{1/p_1}(E|\xi_2|^{q_1})^{1/q_1}.$$

Proof. See Lemma 1.2 in Wang et al. (2009).

LEMMA 3 Let $\{\eta_j : j \geq 1\}$ be a ϕ -mixing sequence with mixing coefficient $\phi(n)$, $\sum_{n=1}^{\infty} \phi^{1/2}(n) < \infty$ and $E\eta_j = 0, j \geq 1$. Suppose that $E|\eta_j|^{r_0} < \infty$ for some $r_0 \geq 2$ and all $j \geq 1$. Then for any $n \geq 1$ and $a \geq 0$,

$$E\left(\sum_{i=a+1}^{a+n} \eta_i\right)^2 \leq C \sum_{i=a+1}^{a+n} E\eta_i^2, \quad (3)$$

$$E\left|\sum_{i=a+1}^{a+n} \eta_i\right|^{r_0} \leq C\left\{\left(\sum_{i=a+1}^{a+n} E\eta_i^2\right)^{r_0/2} + \sum_{i=a+1}^{a+n} E|\eta_i|^{r_0}\right\}. \quad (4)$$

Proof. (3) comes from Theorem 2.1 in Wang et al. (2009). From $\max_{1 \leq j \leq n} |\sum_{i=a+1}^{a+j} \eta_i|^{r_0} \geq |\sum_{i=a+1}^{a+n} \eta_i|^{r_0}$, $\max_{1 \leq i \leq n} |\eta_i|^{r_0} \leq \sum_{i=a+1}^{a+n} |\eta_i|^{r_0}$, (3) and Lemma 1.3 in Wang et al. (2009), we obtain (4).

LEMMA 4 Let $\{\eta_i, i \geq 1\}$ be a ϕ -mixing process and $\mathcal{F}_s^t$ denote the σ -algebra generated by $\{\eta_i, s \leq i \leq t\}$ for $s \leq t$. Suppose that $\{\xi_i, 1 \leq i \leq n\}$ are $\mathcal{F}_{i_1}^{j_1}, \dots, \mathcal{F}_{i_n}^{j_n}$ measurable, respectively with $1 \leq i_1 < j_1 < i_2 < j_2 < \dots < i_n < j_n$, $i_{l+1} - j_l \geq m$, and $|\xi_l| \leq 1$ for $l = 1, \dots, n$. Then,

$$|E(\prod_{l=1}^n \xi_l) - \prod_{l=1}^n E(\xi_l)| \leq 8(n-1)\phi(m). \quad (5)$$

Proof. See Lemma 1.1 in Volkonskii and Rozanov (1959). This result is proved in the case of α -mixing, which also holds true for ϕ -mixing. Note that (5) is also true for complex random variables $\{\xi_i, 1 \leq i \leq n\}$ with $|a|$ replaced by the norm of a complex number a .

LEMMA 5 Under the conditions of Theorem 1, as $n \rightarrow \infty$,

$$(nh)\text{Var}(f_n(x)) = \sigma^2(x) + o(1), \quad (6)$$

$$\sqrt{nh}\{f_n(x) - Ef_n(x)\} \xrightarrow{d} N(0, \sigma^2(x)), \quad (7)$$

where $\sigma^2(x) = f(x) \int K^2(u) du$.

Remark 4. In this lemma, the asymptotic variance of $f_n(x)$ is obtained. In Lin (1983), it is shown, under some regularity conditions, that

$$\frac{f_n(x) - Ef_n(x)}{\text{Var}\{f_n(x)\}} \xrightarrow{d} N(0, 1),$$

where the asymptotic variance of $f_n(x)$ is not given out.

Proof of Lemma 5. We employ so-called small-block and large-block arguments. Let $Z_{ni} = K_h(x - X_i) - EK_h(x - X_i)$, $1 \leq i \leq n$, and

$$S_n = \sqrt{nh}\{f_n(x) - Ef_n(x)\}.$$

Then S_n can be split as $S_n = S'_n + S''_n + S'''_n$, where $S'_n = \sum_{m=1}^k e_{nm}$, $S''_n = \sum_{m=1}^k e'_{nm}$ and $S'''_n = e'_{n,k+1}$, with $e_{nm} = \frac{1}{\sqrt{nh}} \sum_{i=r_m}^{r_{m+1}-1} Z_{ni}$, $e'_{nm} = \frac{1}{\sqrt{nh}} \sum_{i=l_m}^{l_{m+1}-1} Z_{ni}$, $e'_{n,k+1} = \frac{1}{\sqrt{nh}} \sum_{i=k(p+q)+1}^n Z_{ni}$, $r_m = (m-1)(p+q) + 1$, $l_m = (m-1)(p+q) + p + 1$, $m = 1, \dots, k$.

(7) follows if we can verify

$$S'_n \xrightarrow{d} N(0, \sigma^2(x)), \quad (8)$$

$$S''_n = o_p(1), \quad (9)$$

and

$$S'''_n = o_p(1). \quad (10)$$

As a preparation, we first show that

$$\text{Var}(S'_n) = \sigma^2(x) + o(1). \quad (11)$$

From $|f_{1,j}(x, y) - f(x)f(y)| \leq C$, we can see that

$$\begin{aligned} |\text{Cov}(Z_{n1}, Z_{n,j+1})| &= h^2 \left| \int_{R^2} K(u)K(v) \{f_{1,j}(x-hu, x-hv) \right. \\ &\quad \left. - f(x-hu)f(x-hv)\} dudv \right| \leq Ch^2, \end{aligned}$$

which leads to

$$\frac{k}{nh} \sum_{1 \leq i < j \leq p} |\text{Cov}(Z_{ni}, Z_{nj})| \leq C(pk/n)ph \rightarrow 0$$

as $pk/n \rightarrow 1$ and $ph \rightarrow 0$. It follows that

$$\sum_{m=1}^k \text{Var}(e_{nm}) = \frac{kp}{nh} \text{Var}(Z_{n1}) + \frac{2k}{nh} \sum_{1 \leq i < j \leq p} \text{Cov}(Z_{ni}, Z_{nj}) \rightarrow \sigma^2(x). \quad (12)$$

By stationarity, it can be shown that

$$\begin{aligned} \sum_{1 \leq i < j \leq k} |\text{Cov}(e_{ni}, e_{nj})| &= \sum_{l=1}^{k-1} (k-l) |\text{Cov}(e_{n1}, e_{n,l+1})| \\ &\leq k \sum_{l=1}^{k-1} |\text{Cov}(e_{n1}, e_{n,l+1})| \leq \frac{kp}{nh} \sum_{l=1}^{k-1} \sum_{s=l(p+q)-p}^{l(p+q)+p} |\text{Cov}(Z_{n1}, Z_{n,s+1})|. \end{aligned}$$

It follows, from Lemma 2, that

$$\sum_{1 \leq i < j \leq k} \text{Cov}(e_{ni}, e_{nj}) \leq \frac{Ckp}{n} \sum_{s=q}^{\infty} \phi^{1/2}(s) \rightarrow 0. \quad (13)$$

(11) follows from (12) and (13).

Similar to above derivations, we have

$$\begin{aligned} \sum_{m=1}^k \text{Var}(e'_{nm}) &= \frac{kq}{nh} \text{Var}(Z_{n1}) + \frac{2k}{nh} \sum_{1 \leq i < j \leq q} \text{Cov}(Z_{ni}, Z_{nj}) \\ &\leq \frac{Ckq}{n} + \frac{Ckq^2h}{n} \rightarrow 0, \end{aligned}$$

and

$$\begin{aligned}
& \sum_{1 \leq i < j \leq k} |\text{Cov}(e'_{ni}, e'_{nj})| = \sum_{l=1}^{k-1} (k-l) |\text{Cov}(e'_{n1}, e'_{n,l+1})| \\
& \leq k \sum_{l=1}^{k-1} |\text{Cov}(e'_{n1}, e'_{n,l+1})| \leq kq \sum_{l=1}^{k-1} \sum_{r=l(p+q)-(q-1)}^{l(p+q)+(q-1)} |\text{Cov}(Z_{n1}, Z_{n,r+1})| \\
& \leq \frac{Ckq}{n} \sum_{s=p}^{\infty} \phi^{1/2}(s) \rightarrow 0.
\end{aligned}$$

It follows that

$$E(S_n'')^2 \rightarrow 0, \quad (14)$$

which implies (9).

We can also show, from stationarity and Lemma 2, that

$$E(S_n''')^2 \leq \frac{C\{n - k(p+q)\}}{nh} EZ_{n1}^2 \leq Cp/n \rightarrow 0, \quad (15)$$

which leads to (10).

From Lemma 4, we have

$$\left| E e^{it \sum_{m=1}^k e_{nm}} - \prod_{m=1}^k E e^{it e_{nm}} \right| \leq Ck\phi(q) \rightarrow 0, \quad (16)$$

which implies that $\{e_{nm}\}$ are asymptotically independent. Suppose that $\{E_{nm}, 1 \leq m \leq k\}$ are independent random variables and the distribution of E_{nm} is the same as that of e_{nm} for $m = 1, \dots, k$. To prove (8), it suffices to show that

$$\sum_{m=1}^k E_{nm} \xrightarrow{d} N(0, \sigma^2(x)). \quad (17)$$

Let $s_n^2 = k\text{Var}(E_{n1})$ and $F_{nm} = E_{nm}/s_n$, $1 \leq m \leq k$. From the proof of (11), we can see that $s_n^2 \rightarrow \sigma^2(x)$. Hence, (17) is equivalent to

$$\sum_{m=1}^k F_{nm} \xrightarrow{d} N(0, 1). \quad (18)$$

By Lemma 3,

$$\begin{aligned}
& kE|F_{n1}|^3 = s_n^{-3} E|e_{n1}|^3 = ks_n^{-3} (nh)^{-3/2} E \left| \sum_{i=r_1}^{r_1+p-1} Z_{ni} \right|^3 \\
& \leq Cks_n^{-3} (nh)^{-3/2} \left[\sum_{i=r_1}^{r_1+p-1} E|Z_{ni}|^3 + \left\{ \sum_{i=r_1}^{r_1+p-1} EZ_{ni}^2 \right\}^{3/2} \right] \\
& \leq Cks_n^{-3} (nh)^{-3/2} \{ph + (ph)^{3/2}\} \leq Cks_n^{-3} (nh)^{-3/2} ph \\
& \leq Cs_n^{-3} (kp/n) (nh)^{-1/2} \rightarrow 0,
\end{aligned}$$

where we have used $ph \rightarrow 0, s_n^2 \rightarrow \sigma^2(x), kp/n \leq 1$ and $nh \rightarrow \infty$. By the Lindeberg central limit theorem, we obtain (18). Then (8) follows.

From (8), (9), (10) and the Cramer-Wold theorem, we obtain (7). (6) follows from (11), (14) and (15). The proof of Lemma 5 is thus complete.

LEMMA 6 Under the conditions of Theorem 1, as $n \rightarrow \infty$,

$$\omega_n = \max_{1 \leq i \leq 2k+1} |\omega_{ni}(\theta)| \leq Cph^{-1}, a.s., \quad (19)$$

$$\sqrt{n^{-1}h} \sum_{i=1}^{2k+1} \omega_{ni}(\theta) \xrightarrow{d} N(0, \sigma^2(x)), \quad (20)$$

$$(n^{-1}h) \sum_{i=1}^{2k+1} \omega_{ni}^2(\theta) = \sigma^2(x) + o_p(1), \quad (21)$$

$$(n^{-1}h)^{3/2} \sum_{i=1}^{2k+1} |\omega_{ni}(\theta)|^3 = o_p(1), \quad (22)$$

where $\sigma^2(x) = f(x) \int K^2(u) du$.

Proof. (19) follows from $|K| \leq C, q \leq cp$ and $n - k(p+q) \leq Cp$. Observe that

$$\sqrt{n^{-1}h} \sum_{i=1}^{2k+1} \omega_{ni}(\theta) = \sqrt{nh} \{f_n(x) - Ef_n(x)\} + \sqrt{nh} \{Ef_n(x) - f(x)\}. \quad (23)$$

Standard arguments in the kernel estimation of a p.d.f. leads to (e.g., Singh, 1977)

$$\sqrt{nh} \{Ef_n(x) - f(x)\} = \sqrt{nh} O(h^r) = \sqrt{nh^{2r+1}} O(1) = o(1), \quad (24)$$

provided $nh^{2r+1} \rightarrow 0$. From Lemma 5, we have

$$\sqrt{nh} \{f_n(x) - Ef_n(x)\} \xrightarrow{d} N(0, \sigma^2(x)). \quad (25)$$

(23) to (25) imply (20).

We now prove (21). Denote

$$T_n = (n^{-1}h) \sum_{i=1}^{2k+1} \omega_{ni}^2(\theta), \epsilon_i = h^{-1} \{K_h(x - X_i) - EK_h(x - X_i)\}, 1 \leq i \leq n.$$

It is clear that

$$\begin{aligned} T_n &= (n^{-1}h) \sum_{m=1}^k \left\{ \sum_{i=r_m}^{r_m+p-1} (h^{-1}K_h(x - X_i) - f(x)) \right\}^2 \\ &\quad + (n^{-1}h) \sum_{m=1}^k \left\{ \sum_{i=l_m}^{l_m+q-1} (h^{-1}K_h(x - X_i) - f(x)) \right\}^2 \\ &\quad + (n^{-1}h) \left\{ \sum_{i=k(p+q)+1}^n (h^{-1}K_h(x - X_i) - f(x)) \right\}^2 = \sum_{j=1}^9 T_{nj}, \end{aligned}$$

where

$$\begin{aligned}
T_{n1} &= (n^{-1}h) \sum_{m=1}^k \left(\sum_{i=r_m}^{r_m+p-1} \varepsilon_i \right)^2, T_{n2} = (n^{-1}h) \sum_{m=1}^k \left(\sum_{i=l_m}^{l_m+q-1} \varepsilon_i \right)^2, \\
T_{n3} &= (n^{-1}h) \left(\sum_{i=k(p+q)+1}^n \varepsilon_i \right)^2, T_{n4} = (n^{-1}h) \sum_{m=1}^k \left\{ \sum_{i=r_m}^{r_m+p-1} \{h^{-1}EK_h(x-X_i) - f(x)\} \right\}^2, \\
T_{n5} &= (n^{-1}h) \sum_{m=1}^k \left\{ \sum_{i=l_m}^{l_m+q-1} \{h^{-1}EK_h(x-X_i) - f(x)\} \right\}^2, \\
T_{n6} &= (n^{-1}h) \left\{ \sum_{i=k(p+q)+1}^n \{h^{-1}EK_h(x-X_i) - f(x)\} \right\}^2, \\
T_{n7} &= 2(n^{-1}h) \sum_{m=1}^k \left(\sum_{i=r_m}^{r_m+p-1} \varepsilon_i \right) \left\{ \sum_{i=r_m}^{r_m+p-1} \{h^{-1}EK_h(x-X_i) - f(x)\} \right\}, \\
T_{n8} &= 2(n^{-1}h) \sum_{m=1}^k \left(\sum_{i=l_m}^{l_m+q-1} \varepsilon_i \right) \left\{ \sum_{i=l_m}^{l_m+q-1} \{h^{-1}EK_h(x-X_i) - f(x)\} \right\}, \\
T_{n9} &= 2(n^{-1}h) \left(\sum_{i=k(p+q)+1}^n \varepsilon_i \right) \left\{ \sum_{i=k(p+q)+1}^n \{h^{-1}EK_h(x-X_i) - f(x)\} \right\}.
\end{aligned}$$

(21) is valid if we can show that

$$T_{n1} = \sigma^2(x) + o_p(1), \quad (26)$$

and

$$\sum_{j=2}^9 T_{nj} = o_p(1). \quad (27)$$

We first prove (26). Note that

$$T_{n1} = \sum_{m=1}^k e_{nm}^2,$$

where $e_{nm} = \sum_{i=r_m}^{r_m+p-1} \frac{1}{\sqrt{nh}} \{K_h(x-X_i) - EK_h(x-X_i)\}$, $1 \leq m \leq k$. From the proof of Lemma 5, we can see that

$$ET_{n1} = \text{Var}\left(\sum_{m=1}^k e_{nm}\right) - 2 \sum_{1 \leq i < j \leq k} \text{Cov}(e_{ni}, e_{nj}) = \sigma^2(x) + o(1). \quad (28)$$

On the other hand, we will show that

$$\text{Var}(T_{n1}) \rightarrow 0. \quad (29)$$

It is clear, from Lemma 1, that $\{e_{nm}^2 - Ee_{nm}^2, 1 \leq m \leq k\}$ is a ϕ -mixing sequence. Then, from Lemma 3, we have

$$\text{Var}(T_{n1}) \leq Ck\text{Var}(e_{n1}^2) \leq CkE(e_{n1}^4).$$

Using Lemma 3 again, we have

$$\begin{aligned} \text{Var}(T_{n1}) &\leq Ck(nh)^{-2} \left[\sum_{i=r_1}^{r_1+p-1} EZ_{ni}^4 + \left\{ \sum_{i=r_1}^{r_1+p-1} EZ_{ni}^2 \right\}^2 \right] \\ &\leq Ck(nh)^{-2} \{ph + (ph)^2\} \leq Ck(nh)^{-2} ph \\ &\leq C(kp/n)(nh)^{-1} \rightarrow 0, \end{aligned}$$

where $Z_{ni} = K_h(x - X_i) - EK_h(x - X_i)$. Thus we have (29). (26) follows from (28) and (29).

From the proof of Lemma 5, one obtains that

$$ET_{nj} = o(1), j = 2, 3.$$

Noting that T_{n2} and T_{n3} are nonnegative, we thus have

$$T_{nj} = o_p(1), j = 2, 3. \quad (30)$$

Similar to the proofs of (4.17) and (4.18) in Qin et al. (2011), one can show that

$$T_{nj} = o_p(1), 4 \leq j \leq 9. \quad (31)$$

(27) follows from (30) and (31). We thus obtain (21).

(22) can be proved similar to the proof of (4.4) in Qin et al. (2011). The proof of Lemma 6 is thus complete.

Proof of Theorem 1. Based on Lemma 6, Theorem 1 can be proved by using the same method as in the proof of Theorem 1 in Qin et al. (2011). We thus omit its proof.

4. Concluding remarks

We have studied the construction of confidence intervals for a probability density function under a ϕ -mixing sample by using the blockwise technique. The fact that the EL intervals are obtained without a secondary variance estimation is advantageous for dependent data. We plan to extend our results to more complex dependent samples.

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